

The influence of drought on flow-ecology relationships in Ozark Highland streams

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Abstract

1. Drought and summer drying can have strong effects on abiotic and biotic components of stream ecosystems. Environmental flow-ecology relationships may be affected by drought and drying, adding further uncertainty to the already complex interaction of flow with other environmental variables, including geomorphology and water quality.
2. Environment–ecology relationships in stream communities in Ozark Highland streams, USA, were examined over two years with contrasting environmental conditions, a drought year (2012) and a flood year (2013). We analysed fish, crayfish and benthic macroinvertebrate assemblages using two different approaches: (1) a multiple regression analysis incorporating predictor variables related to habitat, water quality, geomorphology and hydrology and (2) a canonical ordination procedure using only hydrologic variables in which forward selection was used to select predictors that were most related to our response variables.
3. Reach-scale habitat quality and geomorphology were found to be the most important influences on community structure, but hydrology was also important, particularly during the flood year. We also found substantial between-year variation in environment–ecology relationships. Some ecological responses differed significantly between drought and flood years, while others remained consistent. We found that magnitude was the most important flow component overall, but that there was a shift in relative importance from low flow metrics during the drought year to average flow metrics during the flood year, and the specific metrics of importance varied markedly between assemblages and years.
4. Findings suggest that understanding temporal variation in flow-ecology relationships may be crucial for resource planning. While some relationships show temporal variation, others are consistent between years. Additionally, different kinds of hydrologic variables can differ greatly in terms of which assemblages they affect and how they affect them. Managers can address this complexity by focusing on relationships that are temporally stable and flow metrics that are consistently important across groups, such as flood frequency and flow variability.

KEYWORDS

aquatic communities, drought, environmental flows, flow-ecology, hydrologic variation

1 | INTRODUCTION

Drought and seasonal drying have strong effects on both biotic and abiotic components of stream ecosystem structure and function (Bond, Lake, & Arthington, 2008; Lake, 2011; Magoulick & Kobza, 2003). Drought acts as an immediate stressor affecting local populations and can also influence the evolution and adaptation of species, local communities and ecosystems on a broad scale (Benejam, Angermeier, Munne, & Garcia-Berthou, 2010; Lytle & Poff, 2004; Matthews & Marsh-Matthews, 2003). Environmental flow assessments have gained increasing momentum in recent years through the development of methods and approaches that help support the management of stream ecosystems (Gillespie, Desmet, Kay, Tilloston, & Brown, 2014; Olden et al., 2014; Poff et al., 2010). There is an implicit assumption that flow-ecology relationships will be temporally stable allowing predictions about how biota will respond to environmental changes (Carlisle, Wolock, & Meador, 2010; Olden et al., 2014). However, it is likely that temporal variation in baseline environmental conditions may play an important role in shaping flow-ecology relationships. Furthermore, global climate change is predicted to alter the hydrologic cycle, resulting in increased intensity, duration, and extent of drought and extreme weather events (Farjad, Gupta, & Marceau, 2015; Hosseinzadeh, Seyedkaboli, Zarei, Akhondali, & Farad, 2015; Masson-Delmotte et al., 2013; Wilhite, Sivakumar, & Pulwarty, 2014; Xenopolous et al., 2005), which might further confound predictions about flow-ecology relationships and management recommendations based on those predictions.

Supra-seasonal drought events are expected to become more common due to global climatic change (Lake, 2000; Xenopolous et al., 2005). Streams in many regions experience a seasonal cycle of drought and flooding that represents the natural disturbance regime in these systems, which plays a crucial role in shaping habitat, diversity and community composition (Lytle & Poff, 2004). The three recognised classes of ecological perturbation are pulse, press and ramp (Bender et al., 1984; Glasby & Underwood, 1996; Lake, 2003). Seasonal stream drying during the summer in regions such as the Ozark Highlands is typical of a press perturbation, in which a disturbance occurs and is maintained at a steady rate (Bender et al., 1984). The same streams may also experience short-term, intense drying events that represent pulse perturbations, in which disturbance occurs and then conditions quickly return to normal levels (Lake, 2003; Lynch & Magoulick, 2016). Supra-seasonal droughts represent extreme ramp disturbances in which the perturbation increases in strength over time (Lake, 2000) and may be much more difficult for organisms to cope with due to a lack of predictability in timing or duration (Lake, 2003; Lynch & Magoulick, 2016). Ideally, natural temporal variation, including infrequent major disturbances such as severe floods (Matthews, 1986; Matthews, Marsh-Matthews, Adams, & Adams, 2014) or supra-seasonal droughts (Lake, 2003), should be integrated into biomonitoring studies (Sousa, 1984; Stoddard, Larsen, Hawkins, Johnson, & Norris, 2006), although this might be challenging due to the logistics of monitoring many sites over

extended periods (MacDonald & Cote, 2014). This kind of extreme temporal variation can confound our understanding of flow-ecology relationships.

Uncertainty, which is commonly associated with our inability to fully characterise the natural spatial and temporal variability inherent in data collections, can further confound our understanding of flow-ecology relationships, especially when trying to account for the complex interactions among variables such as habitat structure, geomorphology and water quality (Poff et al., 2010). The role of these factors in shaping the structure and function of riverine ecosystems is well established (Dudgeon et al., 2006; Poff et al., 1997), but separating out the relative importance of these interactions remains a challenge in the establishment of regional flow standards (Olden et al., 2014; Poff et al., 2010). Our primary objective was to further the understanding of environment-ecology relationships for fish, crayfish and benthic macroinvertebrate communities in Ozark Highland streams between two years with contrasting environmental conditions, a drought year (2012) and a flood year (2013). We hypothesised (1) that there would be temporal variation in environment-ecology relationships between the two years (a drought year versus a flood year) and (2) that flow-ecology relationships would be weaker during the drought year, as reduced flow would lead to an increase in biotic effects relative to abiotic effects. In addition, we sought to better account for the effects of interactions among explanatory variables on the development of flow-ecology relationships.

To address our objectives, we (1) used multiple regression analysis to examine environment-ecology relationships using a best subsets approach that helped us identify the best subset of predictor variables, and (2) we applied a canonical ordination procedure on a larger set of only hydrologic variables to select hydrologic predictors that were most related to our response variables. While previous studies in the region have addressed how aspects of water quality, geomorphology and habitat are related to the distribution and structure of aquatic communities (Matthews, Hough, & Robison, 1992; Matthews & Robison, 1988), we greatly expand on these relationships by integrating newly available flow classification techniques and detailed hydrologic metrics into our modelling methodology (Archfield, Kennen, Carlisle, & Wolock, 2013; Olden et al., 2014; Poff et al., 2010).

1.1 | Study area

The Ozark Highlands (Figure 1) is a Level III Ecoregion (Omernik & Griffith, 2014) stretching across portions of four states in the Interior Highlands of the central United States. It is a crucial centre of biodiversity and endemism for many taxonomic groups of aquatic organisms (TNC-OEAT, 2003). Stream drying is a critical part of the disturbance regime in this ecoregion and has strong effects on many aspects of stream community structure and function, including density and richness of fish assemblages (Dekar & Magoulick, 2007), crayfish habitat selection (DiStefano, Magoulick, Imhoff, & Larson, 2009), seasonal migration and refuge use by fish species (Hodges & Magoulick, 2011), leaf breakdown (Magoulick, 2014) and periphyton

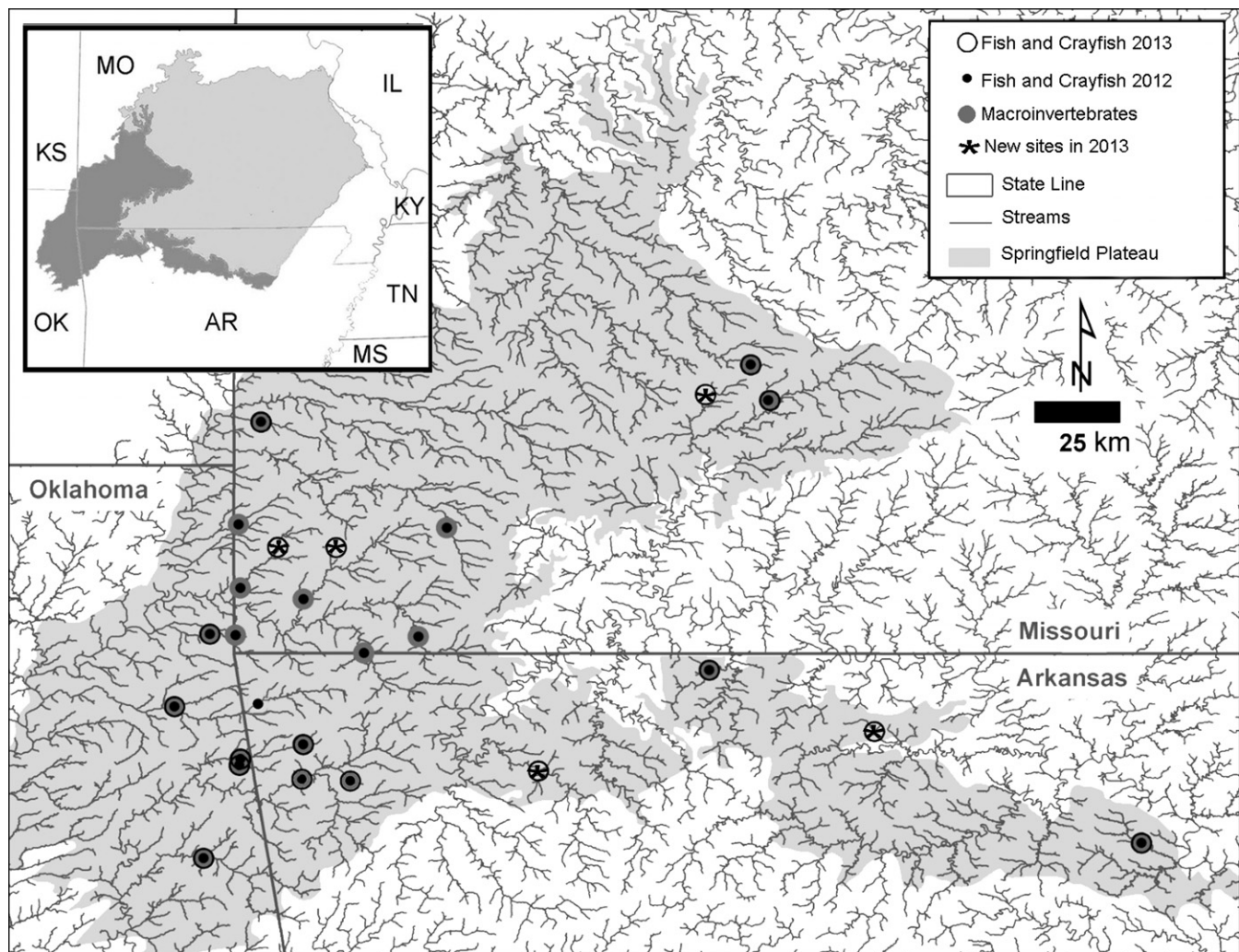


FIGURE 1 Map of study area showing sample sites, stream network and Springfield Plateau (shaded grey). Inset shows location of the Ozark Highlands Level III Ecoregion (light grey), including the Springfield Plateau physiographic province (dark grey) in the Central United States

growth (Lynch & Magoulick, 2016). Freshwater biodiversity in this region is threatened by many anthropogenic impacts, including rapidly growing urban areas and extensive agricultural development affecting water quality (Haggard, 2010; Petersen et al., 2005; Scott, Haggard, Sharpley, & Romeis, 2011), expansion in natural gas extraction (Johnson et al., 2015), displacement of native crayfish by invasive species (Larson, Magoulick, Turner, & Laycock, 2009; Magoulick & DiStefano, 2007) and direct hydrologic alteration of streams via construction of dams and reservoirs and catchment development (The Nature Conservancy, Ozarks Ecoregional Assessment Team (TNC-OEAT), 2003).

2 | METHODS

2.1 | Site selection

Aquatic community sampling was conducted at 26 sampling sites over two summer field seasons (May–July) during 2012 and 2013 in the Springfield Plateau of the Ozark Highlands in north-west

Arkansas, south-west Missouri and north-east Oklahoma, USA (Figure 1). The two years presented a strong contrast in flow conditions throughout the study area: an extreme drought as measured on the Palmer Drought Severity Index (Palmer, 1965) was present for most of summer 2012, whereas sustained higher than normal precipitation resulting in summer flooding characterised many of the sites during summer 2013 (NOAA, 2015; Figure 2).

To facilitate ecological comparisons, all sites were selected within a single ecoregion, the Ozark Highlands; a single physiographic region, the Springfield Plateau; and within a single flow regime, Groundwater Flashy streams, based on the classification of Ozark streams into seven hydrologic flow regimes (Leasure, Magoulick, & Longing, 2016). Groundwater Flashy streams are the most common flow regime in the Ozark Highlands and are characterised by less daily flow variability than any runoff-dominated streams, but less constancy and lower base flow than other groundwater flow regimes. Like other groundwater-influenced flow regimes, they never dry up completely (Leasure et al., 2016). Study streams selected ranged in drainage area from 16 to 542 km² and were located in three

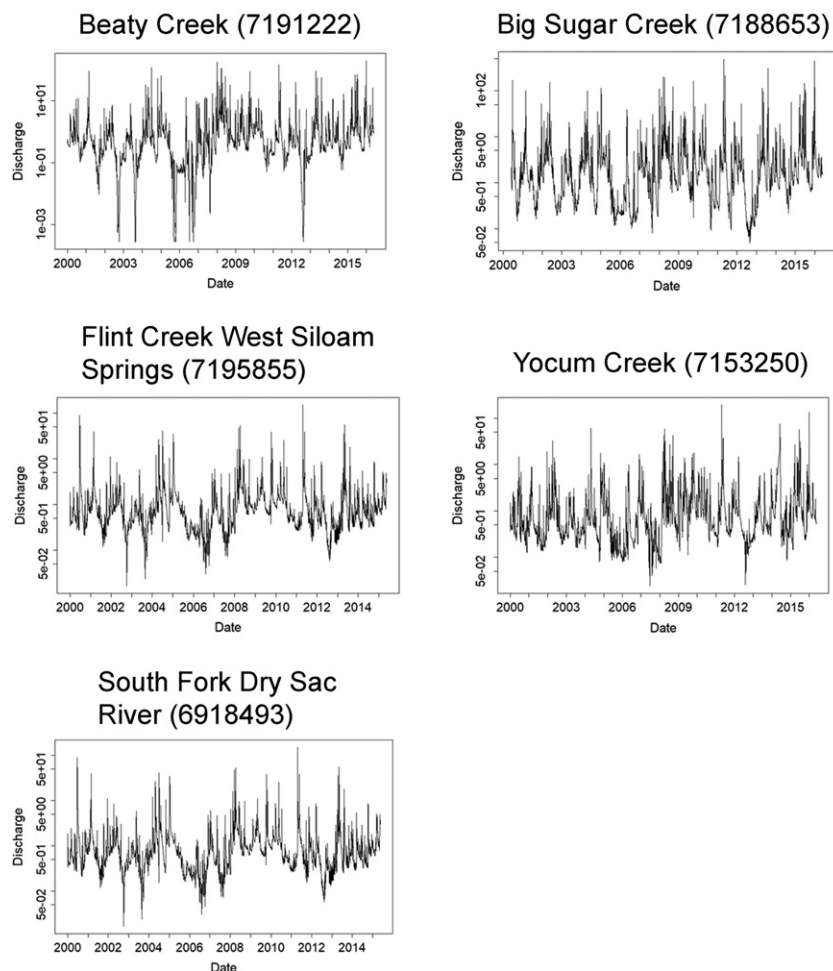


FIGURE 2 Hydrographs from representative sites in the five major drainages in the study area: Arkansas (Neosho, Elk and Illinois River Systems), White and Missouri River Basins with USGS gage ID numbers. Discharge in cubic metres per second (m^3/s). Gap in hydrology data for gage 6,918,493 was due to USGS gage disruption

different major river basins: the Arkansas River basin, White River basin and Osage River basin.

The majority of sites ($n = 20$) were located at USGS stream gages where hydrologic data could be obtained, but ungaged sites ($n = 6$) were also included. All available gaged sites within the flow class and physiographic province were selected. Ungaged site selection was based on a flow regime map of the Ozark Highlands (Leasure et al., 2016); sites were randomly selected from stream segments within the Groundwater Flashy flow regime class. Natural flow conditions were predicted for all sites based on 171 flow metrics comprising five major flow components, magnitude (M), frequency (F), duration (D), timing (T) and rate of change (R) (Leasure et al., 2016). Due to extreme differences in sampling conditions between the two years (drought in 2012 versus extensive flooding in 2013), we were unable to resample eight of the largest sites from the first field season for fish and crayfish during the second season, but added five new sites in 2013 following the same site selection criteria. Twenty-one sites were sampled for fish and crayfish in 2012, 18 in 2013 with 13 overlapping sites between the two years. We sampled macroinvertebrates at 20 sites with total overlap between years. Water quality, habitat and geomorphic measurements were made at all sites in both years. It is important to note that the lack of complete overlap in fish and crayfish sites in 2012

and 2013 limits the strength of the analysis, thereby weakening any conclusions that can be drawn about temporal variation in fish and crayfish communities between years.

2.2 | Habitat, geomorphology and water quality assessment

Habitat variables including wetted width, current velocity, depth, substrate composition and canopy cover were measured. Length and wetted width of each habitat unit were measured with a tape measure, length at the middle of the stream and width at a minimum of five transects along each habitat unit. For habitat units exceeding 10 m in length, an additional transect was added for each additional 5 m. At five evenly spaced points along each transect, stream depth was measured with a top setting wading rod and current velocity was measured at 60% of depth with a Marsh McBirney Flow-Mate 2000 Portable Flowmeter. Substrate size was assessed and recorded using a modified Wentworth scale ranging from 1 (silt) to 7 (bed-rock). Mean depth, mean wetted width and unit length were used to calculate the total volume of each habitat unit, which in turn was used to derive density estimates of fish and crayfish. For richest-targeted habitat (RTH) invertebrate samples, habitat metrics were calculated based only on measurements in the riffle habitat units; for

TABLE 1 Predictor variables used in multiple regression analyses

Predictor variables		
Rapid Habitat Assessment (RHA)	Hydrology 1 (PCA Axis 1; I2012, F2012, C2012)	Water quality (PCA Axis 1; I2013)
Epifaunal substrate/available cover	+MA1—mean daily flow	+Total P
Embeddedness	+DL4—magnitude of minimum annual flow	+Total N
Velocity/depth patterns	+RA3—fall rate	Water quality (PCA Axis 1; I2012, F2012, C2012)
Sediment deposition	—MA4—variability in daily flows	+Total P
Channel flow status	Hydrology 2 (PCA Axis 2; I2012, F2012, C2012)	+Total N
Channel alteration	+MH13—variability across maximum monthly flows	+Conductivity
Frequency of riffles	+FH10—flood frequency	Geomorphology (PCA Axis 1; 2012)
Bank stability	+MH19—skewness in annual maximum flows	+Slope score (1–4)
Bank vegetative protection	Hydrology 1 (PCA Axis 1; I2013, F2013, C2013)	+Maximum bankfull depth (m)
Riparian buffer width	+MH19—skewness in annual maximum flows	+Mean erosion (m)
Hydrologic Disturbance Index (HDI)	+MH13—variability across maximum monthly flows	—Mean bank substrate size
Dam density	+FH10—flood frequency	—Bank Texture—Wentworth Scale (1–7)
Dam storage	+MA45—skewness in annual flows	—Incision ratio
Artificial canals	Hydrology 2 (PCA Axis 2; I2013, F2013, C2013)	—Bankfull substrate size
Road density	+FL2—variability in low flood pulse count	Geomorphology (PCA Axis 1; 2013)
Pollutant discharge sites	+TH3—seasonal predictability of non-flooding	+Mean erosion (m)
Water withdrawals	—FH2—variability in high flood pulse count	—Bank Texture—Wentworth Scale (1–7)
Land cover fragmentation	—FH11—flood frequency	—Bankfull substrate size
	Water Quality (PCA Axis 1; F2013, C2013)	—Mean bank substrate size
	—Total P	
	—Total N	

Range of possible scores given in parentheses for variables with qualitative scores. Specific geomorphology, Water Quality and hydrology metrics vary depending on year and taxa. Taxonomic group (I = macroinvertebrates, F = fish, C = crayfish), and year are indicated for some predictor variables.

qualitative multiple habitat (QMH) samples, measurements taken across all habitat units were combined for the entire reach and used.

Stream geomorphology was assessed at each reach using a protocol specifically designed for high-gradient streams (Willard et al., 2004); measurements included mean bankfull width, mean and maximum bankfull depth, low bank height, ratio of bankfull width to depth, incision ratio, bank slope score (from gentle to extra-steep), wetted width substrate size, bankfull width substrate size, bank substrate size, bank cohesiveness and erosion (Table 1).

A qualitative Rapid Habitat Assessment (RHA) score based on 10 habitat metrics was given to each site. The 10 RHA metrics were specifically designed for the assessment of high-gradient streams and consisted of epifaunal substrate and available cover, embeddedness, velocity/depth patterns, sediment deposition, channel flow status, channel alteration, frequency of riffles, bank stability, bank vegetative protection and riparian vegetative buffer width (Table 1). Each parameter was ranked from 1 (low quality) to 20 (high quality) and tallied for a maximum possible score of 200 (Willard et al., 2004).

Water samples were acquired at each sampling site three times during each sampling year (i.e. spring, summer and winter). Samples were collected using a standard dip method at the thalweg of the stream and stored in a 500-ml Nalgene™ sampling bottle. Prior to sampling, all collection bottles were first rinsed by filling and

emptying with stream water three times. Once the sample was collected, it was stored on ice in a cooler for transport to the laboratory (USEPA, 2007). Water was analysed by the Arkansas Water Resources Center and water quality parameters including chloride, conductivity, fluoride, nitrate, soluble reactive phosphorus (P), sulphate, total nitrogen (N), total P, total suspended solids and turbidity were estimated based on established laboratory procedures. Additionally, at the time of biological sampling, field parameters including temperature, pH, dissolved oxygen, conductivity and salinity were collected using a Hydrolab Model MS5 multiparameter sonde.

2.3 | Hydrologic variable estimation

Sixty-four reference gages identified in the Interior Highlands region (Leasure et al., 2016) were used to calculate a set of 171 flow metrics using the R package *EflowStats* (<https://github.com/USGS-R/EflowStats>; Archfield et al., 2013). Random forest (RF) models were used to predict the value of flow metrics using reference gage flow metrics as response variables. We initially assessed 249 variables describing climate, topography, soil characteristics, geology, ground water, hydrology and land cover (Leasure et al., 2016). A reduced RF model was built for each flow metric including only the 30 most important predictor variables (see table 1 in Leasure et al., 2016).

The reduced model was then used to predict values of each flow metric under natural conditions based on the landscape and climate predictor variables for all stream sites used in the present study. The median of the predicted distribution was used as the value for each flow metric under natural conditions.

2.4 | Aquatic community assessment

At each site, we sampled fish, crayfish and benthic macroinvertebrates. Sampling was stratified by habitat to include three units each of riffles, pools and runs, for a total of nine habitat units per reach. Riffle units ranged from 0.30 to 13.13 m³, pool units from 1.76 to 58.42 m³ and runs from 0.61 to 31.86 m³. Total volume sampled at sites ranged from 21.60 to 199.96 m³. All habitat units were at least 100 m away from road crossings to avoid the hydrologic influence of bridge abutments, culverts and other man-made structures influencing physical stream habitat characteristics (Barbour, Gerritsen, Snyder, & Stribling, 1999).

Fish and crayfish were collected using a Smith-Root Model LR-24 backpack electrofishing unit which has been shown to be effective for sampling fish and crayfish assemblages in Ozark streams (Dauwalter & Pert, 2003; Rabeni, Collier, Parkyn, & Hicks, 1997). Standard LR-24 settings for power output based on ambient stream conductivity were used. Prior to sampling, 1.6-cm² mesh block-nets were placed at the end of each habitat unit to prevent animals from escaping or biasing sampling data by moving from one unit to another. A four-person team conducted three upstream passes per habitat unit. Collections from each pass were kept in separate buckets (one for fish, one for crayfish) until all passes were completed. Each pass was processed separately, and all specimens were identified to species and released live back into the stream. All electrofishing crew members were trained in fish collection techniques and fish were identified to species by the senior author who is highly experienced with the fish fauna of Ozark Highland streams.

Benthic macroinvertebrates were sampled using two methodologies developed for the U.S. Geological Survey National Water Quality Assessment (NAWQA) programme (Moulton, Kennen, Goldstein, & Hambrook, 2002): (1) a semiquantitative richest-targeted habitat (RTH) method and (2) a qualitative multiple habitat (QMH) method. Invertebrate samples were typically collected at sampling sites a few hours prior to fish sampling. A quantitative RTH sample of invertebrates was acquired from riffles using a 500- μ m mesh Slack sampler (50 cm $\times$ 30 cm) equipped with 0.25-m² area PVC frame attached to the upstream end of the sampler. The Slack sampler was positioned immediately downstream of the chosen quadrat and perpendicular to the direction of flow. Large cobble and woody debris were lightly brushed, inspected for any remaining invertebrates and then removed from the sampling area. The sampling area was then agitated by disturbing the substrate upstream of the mouth of the sampler allowing the dislodged invertebrates to flow into the trailing net. The nine discrete subsamples were then composited into a 19-L plastic bucket for processing. Processing consisted of rinsing and

removal of large organic debris, followed by elutriation and sieving (with a 500- μ m sieve) to separate invertebrates and organic debris from inorganic material.

The QMH method was used to document the richness of invertebrate taxa present in all habitat types throughout the sampling reach (Moulton et al., 2002). Crew members assessed the entire sampling reach to determine the number of different instream habitat types present and to estimate the proportion of each type. QMH samples were collected using a D-frame kicknet with 500- μ m mesh. Each habitat type was sampled in proportion to total habitat area for a standardised time of 1 hr per reach. Samples were processed in the field as described for the RTH method.

In the laboratory, RTH and QMH invertebrate samples were sorted on a square gridded subsampling frame of 25 $\times$ 5 cm squares using a fixed-count approach targeting a minimum of 300 organisms (Barbour et al., 1999; Moulton, Carter, Grotheer, Cuffney, & Short, 2000). After pouring the sample into the frame and allowing it to settle evenly, an initial inspection was performed to remove large and rare organisms likely to be missed during subsampling. A grid square was randomly selected, and all organisms present were removed from the grid and processed. Subsampling proceeded in this fashion until a minimum of 300 organisms were counted, with the square in which the 300th organism was counted was also fully processed. All macroinvertebrates were identified to the lowest practical taxonomic level, generally family or genus. To estimate total numbers of organisms, a laboratory subsampling correction factor was used (Moulton et al., 2000) in which the total number of grids was divided by the number of grids sorted during subsampling and multiplied by the number of organisms subsampled. Large and rare organisms taken from the sample as a whole were added to these numbers without a correction factor. Invertebrate community response metrics were then calculated based on these numbers.

2.5 | Biological response metric selection

We calculated biological response metrics for fish, crayfish and macroinvertebrate assemblages. A subset of metrics considered to be most ecologically relevant were chosen for analysis based on published relationships and best professional judgement. Five response metrics were chosen for fish, three for crayfish and five each for RTH and QMH invertebrate samples. We targeted biological response metrics that have been shown to be related to habitat, geomorphology, water quality or hydrology (Barbour et al., 1999; Karr, 1991; Matthews & Robison, 1988; Matthews et al., 1992). Mean values and standard errors for biological response metrics used for all assemblages are summarised in Table 2.

For fish, the five biological response metrics chosen were species richness, Simpson's diversity, total fish density (per volume sampled), per cent centrarchids and per cent species categorised as intolerant in an index of biotic integrity developed for Ozark Highlands fish assemblages (Dauwalter, Pert, & Keith, 2003; Table 2). Per cent centrarchids was chosen as a response metric because most Ozark

centrarchids are considered ecologically tolerant habitat generalists (Dauwalter et al., 2003; Robison & Buchanan, 1988).

For crayfish, the three biological response metrics chosen were Simpson's diversity, total crayfish density per volume sampled and per cent contribution of species designated as habitat generalists in an assessment of invasion risks of crayfish in the eastern United States (Larson & Olden, 2010; Table 2). These are represented by two of the nine species in our dataset, *Faxonius neglectus neglectus* and *Faxonius virilis*. Species richness was not used as a community response metric for crayfish due to the generally low and relatively uniform richness across sites. Species of fish and crayfish collected are listed in Table S1.

For benthic macroinvertebrates, the five biological response metrics chosen were abundance, taxa richness, Simpson's diversity, per cent contribution of individuals belonging to Orders Ephemeroptera, Plecoptera and Trichoptera (EPT) and per cent contribution of family Chironomidae (Table 2). Response metrics calculated from RTH and QMH samples were analysed separately.

2.6 | Data analysis

We examined environment–ecology relationships among assemblages with two approaches: (1) a multiple regression analysis incorporating predictor variables related to habitat, water quality, geomorphology and hydrology and (2) a canonical ordination procedure using only hydrologic variables in which forward selection procedures were used to select predictors that were most related to our response variables.

2.7 | Multiple regression analysis

Multiple regression was used to examine environment–ecology relationships using a “best subsets” approach. This involved first identifying all of the possible multiple regression models derived from all of the possible combinations of the candidate predictors and choosing the models that did the “best” at meeting our pre-defined criteria in each variable category. After further model refinement, the best set of predictor variables was chosen. These analyses were designed to assess the relative importance of hydrology in relation to other categories of predictors, including habitat, geomorphology, catchment-scale disturbance and water quality. Using this approach, we selected predictor variables from each of the predictor categories.

Principal components analysis (PCA) was used as a variable reduction technique for geomorphology, water quality and hydrology. For each predictor variable category, a Pearson correlation matrix was inspected to ensure that there were no strong correlations ($r \geq .70$) between predictors. Those that exceeded this criterion were dropped from further analysis. Eleven geomorphological variables (Table S2) were reduced to a single surrogate variable (i.e. PCA Axis 1). A similar procedure was carried out for water quality with a reduction from 4 to 1 surrogate variable. For hydrology, 14 variables accounting for the magnitude, frequency, duration, timing and rate of change of flow were reduced to two surrogate predictor variables (i.e. PCA axes 1 and 2, herein referred to as Hydrology 1 and Hydrology 2, respectively). These two surrogate hydrologic predictor variables were chosen because combined they explained >50% of

TABLE 2 Mean ($\pm$ SE) values for biological response metrics used in multiple regression analysis in 2012 and 2013 and results of paired *t* tests between years

Variable	2012	2013	<i>t</i>	<i>df</i>	<i>p</i> -value
Fish species richness	16 ($\pm$ 0.17)	15 ($\pm$ 0.19)	2.00	13	.067
Fish Simpson's diversity	0.73 ($\pm$ 0.01)	0.75 ($\pm$ 0.01)	−1.59	13	.137
Fish % intolerant	0.70 ($\pm$ 0.01)	0.68 ($\pm$ 0.01)	0.62	13	.623
Fish % Centrarchidae	0.03 ($\pm$ 0.01)	0.05 ($\pm$ 0.01)	−1.94	13	.075
Fish total density	11.00 ($\pm$ 0.25)	15.54 ($\pm$ 0.69)	−1.12	13	.284
Crayfish simpson's diversity	0.21 ($\pm$ 0.01)	0.30 ($\pm$ 0.01)	0.86	8	.416
Crayfish % generalist	0.75 ($\pm$ 0.02)	0.44 ($\pm$ 0.02)	1.96	8	.079
Crayfish total density	3.68 ($\pm$ 0.19)	7.30 ($\pm$ 0.38)	−2.20	8	.053
RTH abundance	2,382 ($\pm$ 625.13)	4,014 ($\pm$ 659.32)	−3.21	19	.005
RTH taxa richness	22 ($\pm$ 1.25)	19 ($\pm$ 1.02)	1.94	19	.068
RTH Simpson's diversity	0.808 ($\pm$ 0.02)	0.776 ($\pm$ 0.02)	1.12	19	.278
RTH % EPT taxa	0.496 ($\pm$ 0.04)	0.599 ($\pm$ 0.042)	−2.12	19	.048
RTH % Chironomidae	0.084 ($\pm$ 0.03)	0.148 ($\pm$ 0.047)	−1.30	19	.209
QMH abundance	2,306 ($\pm$ 662.40)	3,067 ($\pm$ 355.42)	−1.25	19	.226
QMH taxa richness	24 ($\pm$ 1.98)	27 ($\pm$ 1.19)	−1.51	19	.147
QMH Simpson's diversity	0.722 ($\pm$ 0.05)	0.834 ($\pm$ 0.02)	−2.01	19	.058
QMH % EPT taxa	0.234 ($\pm$ 0.05)	0.352 ($\pm$ 0.04)	−2.80	19	.011
QMH % Chironomidae	0.069 ($\pm$ 0.03)	0.189 ($\pm$ 0.04)	−5.29	19	<.005

Significant values in bold. RTH, richest-targeted habitat Invertebrate Samples; QMH, qualitative multi-habitat Invertebrate Samples.

the variation and loaded strongly on different flow metrics across the two PCA axes.

Sites used in this study encompassed a wide hydrologic disturbance gradient of conditions ranging from reference quality to highly disturbed based on a Hydrologic Disturbance Index (HDI; Falcone, Carlisle, & Weber, 2010). Selected sites ranged from an HDI score of 6 (least disturbed) to 29 (most disturbed) with a mean HDI score of 13.7 and standard error of 1.22. This HDI consisted of seven catchment-scale metrics of disturbance, including presence of major dams; change in reservoir storage from 1950 to 2009; percentage of streamlines coded as canals, ditches or pipelines; road density; distance to nearest major pollutant discharge site; freshwater withdrawal estimates; and fragmentation of undeveloped land in the catchment (Falcone et al., 2010). Because HDI is a composite of catchment-level attributes related to land use, it is not specifically considered a hydrologic variable in the sense that Hydrology 1 and Hydrology 2 are. The HDI and RHA scores were included as predictor variables in our multiple regression analyses after a Pearson correlation matrix was inspected to ensure that there was no strong correlation ($r \geq .70$) between them. Both of these indices contain a number of ecologically relevant metrics but represent two distinctly different spatial scales, the catchment and the reach, respectively (Falcone et al., 2010; Willard et al., 2004).

All variables were graphically examined using box-plots and histograms to check for outliers and normality of distributions. Variables were transformed as needed to improve normality and to meet the assumptions of analyses. This was done separately for the 2012 and 2013 datasets. Paired *t* tests were performed for between-year differences (2012 and 2013) in response variable means.

Multiple regression analyses were performed in SYSTAT 13, and models were ranked using Akaike information criteria corrected for small sample size (AICc). We used *p*-values to determine whether parameter estimates differed from 0. Delta AICc values were calculated, and in cases where these values were within 2 points of the best models, those models were considered equally valid and are reported. Residual plots were visually inspected to ensure the assumption of homogeneity of variance was met.

2.8 | Redundancy analysis

Redundancy analysis (RDA) was used to evaluate response variable-hydrology relationships for sampling years and methods separately. Predicted rather than observed hydrology was used for gaged and ungaged sites in the RDA. To reduce bias and increase precision and accuracy of variable selection, Leasure et al.'s (2016) selection criteria were used to reduce the initial 171 hydrologic variables to 154 prior to the forward selection process. Forward selection in CANOCO 4.5 was then used to select hydrologic variables that were related to response variables. Only those hydrologic variables with $\lambda \geq 0.07$ after entry into the RDA model were retained.

Prior to RDA, response variables were centred and standardised. For this analysis, scaling of ordination scores focused on inter-response variable correlation rather than intersample distance;

therefore, the response variable scores were standardised to prevent variables with large variances from disproportionately influencing ordination diagrams (ter Braak & Smilauer, 1998). Monte Carlo permutations were performed for each RDA to test the significance of the canonical axes together and to determine the overall importance of remaining hydrologic variables. Values and definitions for all significant hydrologic variables in the RDAs are presented in Table 3.

3 | RESULTS

3.1 | Predictor variable PCAs and paired *t* tests

3.1.1 | Fish and crayfish

In 2012, Hydrology 1 (PCA Axis 1) loaded positively on MA1, DL4 and RA3 and negatively on MA4 and Hydrology 2 (PCA Axis 2) loaded positively on MH13, FH10 and MH19 (Table 1). In 2013, Hydrology 1 loaded positively on MH19, MH13, FH10 and MA45, whereas Hydrology 2 loaded positively on FL2 and TH3 and negatively on FH2 and FH11 (Table 1). In 2012, Hydrology 1 explained 30.7% variance and Hydrology 2 explained 20.8% variance, whereas in 2013, Hydrology 1 explained 27.5% variance and Hydrology 2 explained 22.8% variance. In 2012, geomorphology (PCA Axis 1) loaded positively on slope, bankfull depth and erosion and negatively on bank substrate, bank texture, incision ratio and bankfull width substrate, whereas in 2013, geomorphology loaded positively on erosion and negatively on bank texture and bankfull substrate. Geomorphology explained 31.8% and 28.2% of the variance in 2012 and 2013, respectively. In 2012, Water Quality (PCA Axis 1) loaded positively on total P, total N and conductivity, whereas in 2013, Water Quality loaded negatively on Total P and Total N (Table 1). Water Quality explained 50.6% and 37.2% of the variance in 2012 and 2013, respectively. Paired *t* tests showed no significant between-year effects on any fish or crayfish response metrics (Table 2).

3.1.2 | Macroinvertebrates

Hydrology 1 (PCA Axis 1) loaded positively on MA1, DL4 and RA3 and negatively on MA4, and Hydrology 2 (PCA Axis 2) loaded positively on MH13, FH10 and MH19 (Table 1). Hydrology 1 explained 30.2% variance and Hydrology 2 explained 20.8% variance. In 2012, geomorphology loaded positively on slope, bankfull depth and erosion and negatively on bank texture, bankfull substrate and bank substrate, whereas in 2013, geomorphology loaded positively on erosion and negatively on bank texture, bankfull substrate and bank substrate. Geomorphology explained 31.0% and 30.2% of the variance in 2012 and 2013, respectively. In 2012, Water Quality loaded positively on total P, total N and conductivity, whereas in 2013, Water Quality loaded positively on total P and total N (Table 1). Water Quality explained 51.5% and 35.0% of the variance in 2012 and 2013, respectively. Paired *t* tests showed significant between-year effects on RTH abundance, RTH per cent EPT taxa, QMH per cent EPT taxa and QMH per cent Chironomidae (Table 2).

TABLE 3 Hydrologic metrics (Olden & Poff, 2003) used in redundancy analysis for 2012 and 2013 with mean plus or minus the standard error ($\pm$ SE)

Code	Definition	Category	Mean ($\pm$ SE)
2012 Fish			
DH17	High flow duration where upper threshold is defined as 7 times median flow	Duration of high flows	27.42 ($\pm$ 0.34)
DL16	Low flow pulse duration	Duration of low flows	13.60 ($\pm$ 0.30)
MA4	Coefficient in variation of logs in daily flows	Magnitude of average flows	126.81 ($\pm$ 2.78)
MA5	Skewness in daily flows	Magnitude of average flows	4.20 ($\pm$ 0.36)
RA40	Variability in fall rate	Rate of change in flows	626.34 ($\pm$ 11.00)
2012 Crayfish			
DL13	Mean of 30-day minima of daily discharge	Duration of low flows	0.24 ($\pm$ 0.02)
DL14	Low exceedance flows (magnitude of flows exceeded 75% of the time divided by median daily flow over all years)	Duration of low flows	0.36 ($\pm$ 0.02)
ML6	Mean minimum June flows	Magnitude of low flows	9.11 ($\pm$ 2.18)
TH1	Julian date of annual maximum	Timing of high flows	94.87 ($\pm$ 2.34)
TH3	Seasonal predictability of non-flooding (maximum proportion of year during which no floods have occurred during period of record)	Timing of high flows	0.21 ($\pm$ 0.01)
2012 RTH invertebrates			
FH3	High flood pulse count (number of annual occurrences with $3\times$ mean daily flow)	Frequency of high flows	79.88 ($\pm$ 2.23)
MA23	Mean monthly December flows	Magnitude of average flows	67.25 ($\pm$ 13.93)
MA25	Coefficient in variation of February flows	Magnitude of average flows	99.82 ($\pm$ 4.25)
MH22	High flow volume (mean of area between hydrograph and upper threshold of $3\times$ median annual flow)	Magnitude of high flows	81.74 ($\pm$ 8.27)
ML10	Mean minimum October flows	Magnitude of low flows	5.98 ($\pm$ 1.66)
2012 RTH invertebrates			
ML14	Mean of annual minimum flows	Magnitude of low flows	0.19 ($\pm$ 0.01)
ML17	Baseflow index (7-day minimum flow divided by mean annual daily flow averaged across all years)	Magnitude of low flows	0.09 ($\pm$ 0.01)
2012 QMH invertebrates			
DH24	Mean annual number of flood free days	Duration of high flows	156.57 ($\pm$ 0.98)
MA7	Range in daily flow (ratio of 20th/80th percentile in daily flow over all years)	Magnitude of average flows	9.96 ($\pm$ 0.74)
MH6	Mean maximum June flows	Magnitude of high flows	274.41 ($\pm$ 68.5)
ML17	Baseflow index (7-day minimum flow divided by mean annual daily flow averaged across all years)	Magnitude of low flows	0.09 ($\pm$ 0.01)
ML18	Variability in baseflow index (coefficient in variation in ML17)	Magnitude of low flows	87.19 ($\pm$ 11.13)
ML14	Mean of annual minimum flows	Magnitude of low flows	0.19 ($\pm$ 0.01)
2013 Fish			
FH5	Flood frequency (mean number of high flow events per year)	Frequency of high flows	7.96 ($\pm$ 0.10)
MA36	Variability across monthly flows 1 (variability in daily flows divided by median monthly flows where variability is calculated as range)	Magnitude of average flows	21.79 ($\pm$ 1.11)
MA39	Variability across monthly flows 2 (coefficient of variation in mean monthly flows)	Magnitude of average flows	132.53 ($\pm$ 1.55)
MH6	Mean maximum June flows	Magnitude of high flows	143.09 ($\pm$ 12.67)
TL3	Seasonal predictability of low flow (proportion of low-flow events $\geq$ 5-year magnitude falling in a 60-day seasonal window)	Timing of low flows	0.18 ($\pm$ <0.01)
2013 Crayfish			
FH5	Flood frequency (mean number of high flow events per year)	Frequency of high flows	7.96 ($\pm$ 0.10)
2013 Crayfish			
MA39	Variability across monthly flows 2 (coefficient of variation in mean monthly flows)	Magnitude of average flows	132.53 ($\pm$ 1.55)
MA41	Mean annual runoff	Magnitude of average flows	0.88 ($\pm$ 0.02)
MH2	Mean maximum February flows	Magnitude of high flows	269.26 ($\pm$ 35.20)

(Continues)

TABLE 3 (Continued)

Code	Definition	Category	Mean ($\pm$ SE)
ML2	Mean minimum February flows	Magnitude of low flows	12.67 ($\pm$ 1.32)
TL3	Seasonal predictability of low flows (proportion of low-flow events $\geq$ 5-year magnitude falling in a 60-day seasonal window)	Timing of low flows	0.18 ($\pm$ <0.01)
2013 RTH invertebrates			
MA38	Variability in monthly flows divided by median monthly flows	Magnitude of average flows	5.66 ($\pm$ 0.25)
MA41	Mean annual runoff (Mean annual flow divided by catchment area)	Magnitude of average flows	0.88 ($\pm$ 0.02)
MA5	Skewness in daily flows (Mean daily flows divided by median daily flows)	Magnitude of average flows	4.06 ($\pm$ 0.33)
MH18	Variability across annual maximum flows	Magnitude of high flows	11.03 ($\pm$ 0.21)
TA2	Predictability of flow composed of two additive components: constancy and contingency (Olden & Poff, 2003)	Timing of average flows	49.23 ($\pm$ 0.04)
DL16	Low flow pulse duration (mean duration of all occurrences during which magnitude of flow remains below the 25th percentile)	Duration of low flows	13.51 ($\pm$ 0.28)
2013 QMH invertebrates			
MA24	Coefficient of variation in January flows	Magnitude of average flows	106.3 ($\pm$ 2.89)
MH14	Mean of annual maximum flows	Magnitude of high flows	115.48 ($\pm$ 11.09)
TL3	Seasonal predictability of low flows (proportion of low-flow events $\geq$ 5-year magnitude falling in a 60-day seasonal window)	Timing of low flows	0.18 ($\pm$ <0.01)

3.2 | Multiple regression analysis

3.2.1 | Fish

Four of five and one of five environment–fish ecology relationships were significant in 2012 and 2013, respectively (Table 4; Figures 3 and 4). In 2012, fish species richness was positively related to RHA, and in 2013, species richness was positively related to Hydrology 1 and Hydrology 2 (Table 4). In 2012, Simpson's diversity was negatively related to HDI; however, none of the models for Simpson's diversity were significant in 2013. In 2012, per cent centrarchids was negatively related to geomorphology; none of the models for per cent centrarchids was significant in 2013. Total density was positively related to RHA and geomorphology in 2012, and total density was positively related to Hydrology 1 and Hydrology 2 in 2013. Per cent intolerant species were not significantly related to any predictor variable in either year.

3.2.2 | Crayfish

In 2012 and 2013, two and zero of three, respectively, environment–crayfish ecology relationships were significant (Table 4; Figure 3). Crayfish diversity was positively related to water quality in 2012, but showed no significant relationships in 2013. Per cent generalist crayfish was positively related to geomorphology in 2012, but showed no significant relationships in 2013. Crayfish density was not significantly related to predictor variables in either year.

3.2.3 | Macroinvertebrates

In 2012 and 2013, two and four, respectively, of the five RTH environment–macroinvertebrate ecology relationships were

significant (Table 4; Figures 3 and 4). Macroinvertebrate abundance showed no significant relationships with environmental predictor variables in either year. Macroinvertebrate taxa richness was negatively related to geomorphology in 2012 and positively related to Hydrology 2 in 2013. Simpson's diversity showed no significant relationships in 2012, but was positively related to Hydrology 2 in 2013. Per cent EPT taxa were negatively related to geomorphology in 2012 and positively related to RHA in 2013. Per cent EPT was also positively related to Water Quality in 2013 as part of the RHA + Water Quality model. Per cent Chironomidae showed no significant relationships in 2012, but was negatively related to RHA in 2013.

In 2012 and 2013, one and four, respectively, of the five QMH environment–macroinvertebrate ecology relationships were significant (Table 4; Figures 3 and 4). In 2012, macroinvertebrate abundance was not significantly related to environmental variables, whereas in 2013, macroinvertebrate abundance was negatively related to RHA and positively related to Hydrology 1. Taxa richness was not related to any environmental variables in 2012, but was negatively related to geomorphology and HDI and positively related to Hydrology 2 in 2013. Simpson's diversity was not related to predictor variables in either year. In both years, per cent EPT was positively related to RHA. In 2012, per cent chironomids was not related to any environmental variables; however, in 2013, per cent chironomids was positively related to HDI.

Rapid Habitat Assessment (RHA) score was the most selected predictor in the multiple regression analysis and comprised all or part of 14 significant multiple regression models (Table 4). Geomorphology was the second most selected predictor variable and was part of nine best models. Hydrology 2 was in six, Hydrologic Disturbance Index (HDI) was in four, Hydrology 1 was in three, and Water Quality was in two significant best models. Overall, there were 14

significant best models in 2012 and 20 significant best models in 2013 (Table 4).

3.3 | Redundancy analysis

3.3.1 | Fish and crayfish

In 2012, fish assemblages were significantly related to hydrologic metrics representing the magnitude of average flow, duration of high and low flow and rate of change (Table 3; Figure 5). In 2012, crayfish assemblages were significantly related to the duration and magnitude of low flow and to the timing of high flow metrics. Hydrologic metrics accounting for the duration and magnitude of flow were the most important overall, with 7 of 10 significant predictors belonging to one of these two flow components. Among all five flow components, metrics accounting for low, high and average flows were of almost equal importance, with four, three and three of each of the significant metrics belonging to low, high and average flows, respectively. No single metric was important to both fish and crayfish assemblages (Table 3; Figure 5).

In 2013, fish assemblages were significantly related to the magnitude of average and high flow, timing of low flow and frequency of high flow metrics (Table 3; Figure 5). Crayfish assemblages were significantly related to magnitude of average, high and low flow, timing of low flow and frequency of high flow metrics. Magnitude was the most important flow component overall, with 7 of 11 significant predictors belonging to this component group. Metrics relating to low, high and average flows were of almost equal importance, with three, four and four of each of the significant metrics belonging to those flow components, respectively (Table 3; Figure 5).

3.3.2 | Macroinvertebrates

In 2012, RTH macroinvertebrate assemblages were significantly related to magnitude of average flow, magnitude of low flow, magnitude of high flow and frequency of high flow (Table 3; Figure 6). QMH assemblages were significantly related to magnitude of average flow, magnitude of low flow, magnitude of high flow and duration of high flow (Table 3; Figure 6). Magnitude was the most important flow component overall, with 11 of 13 significant predictors between the two sampling methods belonging to this component group. Particularly important were metrics related to magnitude of low flows, which made up more than half of those significant magnitude metrics. Two metrics, ML14 and ML17, were important in both RTH and QMH assemblages (Table 3; Figure 6).

In 2013, RTH macroinvertebrate assemblages were significantly related to magnitude of average flow, magnitude of high flow, duration of low flow and timing of average flow (Table 3; Figure 6). QMH assemblages were significantly related to magnitude of average flow, magnitude of high flow and timing of low (Table 3; Figure 6). Magnitude was again the most important flow component overall, with 6 of 9 significant predictors. There was, however, a shift in importance from metrics related to magnitude of low flows

to magnitude of average flows. Metrics related to magnitude of average flows comprised 4 of the 6 significant magnitude metrics in 2013 (Table 3).

Considering all taxa with respect to average, low and high flow conditions, metrics related to low flow were relatively more important during the drought year, with 43% of all metrics being low-flow-related in 2012, while only 25% were low-flow-related in 2013 (Table 3). Average flow conditions appeared to be more important in the flood year, with 45% of important metrics being average-flow-related in 2013, while only 26% were average flow-related in 2012. Metrics relating to high flow remained consistent at 30% between the two years (Table 3). With respect to flow components, magnitude metrics were the most important across both years at 61% in the drought year and 65% in the flood year. Duration was the second most important flow component in the drought year at 22%, and timing was second most important in the flood year at 20%.

4 | DISCUSSION

We found substantial differences between 2012 (drought year) and 2013 (flood year) in flow-ecology relationships, in support of our primary hypothesis. Although our second hypothesis was supported by weaker statistical relationships during the drought year for the macroinvertebrate assemblages, the opposite was true for fish and crayfish, indicating that changes in annual drought/flood conditions can result in very different effects on the taxonomic groups that comprise stream communities.

The most consistently selected predictor variables across assemblages and years in the MLR analysis were Rapid Habitat Assessment (RHA) score and geomorphology, indicating that reach-scale habitat factors were very important in maintaining the community condition of these streams. This also may indicate that habitat quality and geomorphological features at this scale may, in part, mitigate the adverse effects of water quality, hydrology and catchment-scale disturbance (Poff et al., 2010). Hydrology also played an important role; when taken together, the Hydrology 1 and Hydrology 2 variables accounted for the same number of best models as geomorphology. One notable difference was that there were no significant best models that included hydrologic variables during the drought year, whereas there were nine best models that included hydrology during the flood year. This may indicate that hydrologic variables have a strong influence on these communities; however, severe drought can potentially reduce their importance in favour of reach and microhabitat scale variables.

We saw a difference in significant predictor variables selected for fish species richness between the drought year (e.g. RHA) versus the flood year (e.g. Hydrology 1 and Hydrology 2). Several reach-scale parameters that comprise the RHA score are related to heterogeneity of habitat (Willard et al., 2004), potentially accounting for a greater number of species utilising sites with a higher RHA score in 2012. Many studies (Gorman & Karr, 1978; Heithaus & Grame, 1997; Schlosser, 1982) have shown that metrics reflecting fish

TABLE 4 Best multiple regression models for 2012 (drought year) and 2013 (flood year) for the response metrics used to assess the relationship between fish, crayfish, and RTH and QMH macroinvertebrate assemblages and habitat, geomorphology, hydrology and water quality

Response variable	Best model	Std. coefficient	R ²	p-value
2012 (drought year)				
Fish				
Fish species richness	RHA	0.432	.187	.050
	Geomorphology	−0.338	.114	.134
	HDI	−0.328	.108	.147
Fish Simpson's Diversity	HDI	−0.487	.237	.025
	HDI + Hydrology 1	−0.473, 0.205	.279	.030, .322
Fish % Intolerant	Hydrology 2	0.327	.107	.148
	HDI	−0.318	.101	.159
	Geomorphology	−0.21	.044	.361
	Water Quality	−0.187	.035	.416
Fish % Centrarchidae	Geomorphology + RHA	−0.483, −0.361	.285	.030, .095
	Geomorphology	−0.402	.162	.071
	Hydrology 1 + Geomorphology	−0.318, −0.350	.26	.140, .106
	Hydrology 1	−0.375	.141	.094
Fish Total Density	RHA + Geomorphology	0.400, 0.789	.64	.013, <.001
	Hydrology 2 + RHA + Geomorphology	0.021, −0.361, −0.471	.286	.932, .104, .076
Crayfish				
Crayfish Simpson's Diversity	Water Quality + Geomorphology	0.529, −0.520	.312	.049, .053
	Water Quality + HDI	0.536, −0.590	.297	.043, .063
	Hydrology 2	0.351	.123	.167
	Hydrology 2 + Water Quality	0.427, 0.387	.268	.089, .119
	Water Quality	0.426	.182	.088
	Geomorphology	−0.291	.085	.257
Crayfish % Generalist	Geomorphology	0.547	.299	.015
	Geomorphology + RHA	0.488, −0.313	.394	.026, .134
Crayfish Total Density	Geomorphology	0.376	.141	.113
	Water Quality	0.335	.112	.161
	Hydrology 2	−0.252	.064	.297
	HDI	0.251	.063	.103
	Hydrology 1 + Geomorphology	−0.258, 0.414	.207	.268, .085
Macroinvertebrates				
RTH Abundance	HDI	0.31	.096	.184
	Hydrology 2	0.285	.07	.258
	RHA	−0.234	.055	.321
	Hydrology 1	0.085	.007	.720
	Geomorphology	0.074	.005	.758
	Water Quality	0.16	.026	.500
	Hydrology 2, HDI	0.234, 0.284	.15	.314, .244
RTH Richness	Geomorphology + Hydrology 1	−0.503, 0.342	.340	.021, .102
	Geomorphology	−0.473	.224	.035

(Continues)

TABLE 4 (Continued)

Response variable	Best model	Std. coefficient	R ²	p-value
RTH Simpsons Diversity	Geomorphology	−0.371	.137	.108
	Geomorphology + Hydrology 1	−0.400, 0.349	.258	.073, .115
	Hydrology 1	0.315	.099	.177
	Geomorphology + HDI	−0.531, 0.298	.200	.055, .263
RTH EPT	Geomorphology	−0.39	.152	.089
	Geomorphology + Hydrology 1	−0.416, 0.308	.246	.066, .163
	Water Quality	−0.274	.075	.243
	Hydrology 1	0.273	.074	.245
	Geomorphology + Hydrology 1 + HDI	−0.615, 0.373, 0.360	.334	.024, .094, .166
RTH Chironomid	Water Quality	0.154	.024	.517
	Hydrology 2	0.151	.023	.647
	HDI	0.128	.016	.592
	Geomorphology	0.08	.006	.737
	RHA	−0.032	.001	.894
	Hydrology 1	0.023	.001	.094
QMH Abundance	HDI	0.335	.112	.148
	RHA	−0.297	.088	.204
	Water Quality	0.233	.054	.323
	Geomorphology	0.211	.044	.373
	Hydrology 1	0.208	.043	.380
	Hydrology 2	−0.016	.000	.947
	HDI, Hydrology 1	0.369, 0.257	.177	.114, .263
	RHA, Hydrology 1	−0.360, 0.288	.167	.131, .222
	RHA, HDI	−0.233, 0.283	.405	.320, .231
QMH Richness	Geomorphology	−0.348	.121	.133
	HDI	−0.295	.087	.207
	Geomorph + Water quality	−0.481, 0.296	.191	.065, .242
	Hydrology 1	−0.191	.037	.42
QMH Simpsons Diversity	RHA	0.372	.139	.106
	Geomorphology	−0.348	.121	.132
QMH EPT	RHA	0.517	.267	.02
	RHA + Hydrology 2	0.525, 0.027	.343	.016, .180
QMH Chironomid	Hydrology 1	0.151	.023	.524
	Hydrology 2	0.141	.02	.555
QMH Chironomid	HDI	0.099	.01	.678
	Geomorphology	0.211	.044	.373
	Water quality	0.233	.054	.323
	RHA	−0.297	.088	.204
2013 (flood year)				
Fish Richness	Hydrology 1 + Hydrology 2	0.434, 0.450	.454	.040, .034
	Hydrology 1 + Hydrology 2 + Geomorphology	0.676, 0.618, 0.438	.546	.011, .010, .114
	Hydrology 2	0.328	.107	.184
Fish Simpson's Diversity	Geomorphology	−0.367	.135	.134
	Hydrology 1	0.306	.093	.217
	RHA	0.257	.066	.303

(Continues)

TABLE 4 (Continued)

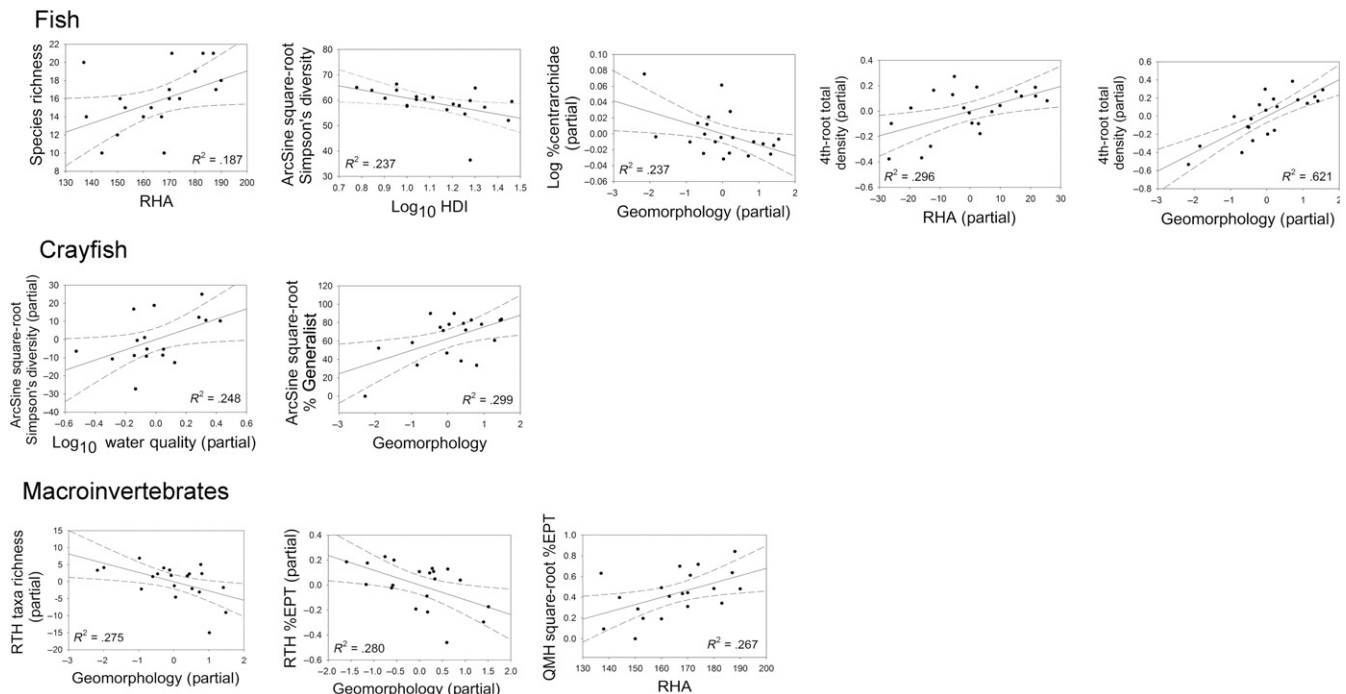
Response variable	Best model	Std. coefficient	R ²	p-value
Fish % Intolerant	Geomorphology	−0.163	.026	.519
	HDI	−0.103	.011	.684
	Water quality	−0.06	.004	.814
	Hydrology 2	−0.033	.001	.897
	Hydrology 1	0.005	.000	.983
	RHA	0.002	.000	.995
Fish % Centrarchidae	Water quality	0.406	.164	.095
Fish Total Density	Water quality	−0.335	.112	.175
	Hydrology 1	0.329	.108	.182
	HDI	−0.295	.087	.234
	HDI + Water quality	−0.333, −0.369	.222	.166, .128
Crayfish				
Crayfish Simpson's Diversity	Water quality	−0.355	.126	.148
	HDI	−0.184	.034	.464
Crayfish Simpson's Diversity	Water quality + HDI	−0.378, −0.223	.176	.129, .358
Crayfish % Generalist	Hydrology 2	−0.459	.211	.055
	RHA	−0.44	.194	.067
	RHA + Hydrology 2	−0.326, −0.354	.306	.172, .140
Crayfish Density	Water quality	−0.368	.135	.133
	Hydrology 2	−0.254	.065	.308
Macroinvertebrates				
RTH Abundance	HDI	0.179	.032	.449
	Hydrology 2	0.177	.031	.456
	Hydrology 1	0.173	.03	.466
	RHA	0.080	.006	.736
	Water quality	0.080	.006	.738
	Geomorphology	0.012	.000	.960
RTH Richness	Hydrology 2	0.415	.172	.069
	Hydrology 2 + HDI	0.453, −0.347	.291	.041, .110
	Geomorphology	−0.34	.116	.142
	Geomorphology + Hydrology 2	−0.254, 0.353	.233	.262, .126
	HDI	−0.296	.088	.205
RTH Simpsons Diversity	Hydrology 2 + Geomorphology	0.429, −0.400	.428	.036, .050
	Hydrology 2 + HDI	0.567, −0.355	.402	.008, .077
	Hydrology 2	0.527	.278	.017
RTH EPT	RHA	0.476	.227	.034
	RHA + Water Quality	0.536, 0.339	.338	.016, .109
	RHA + Hydrology 2 + Water Quality	0.547, −0.312, 0.367	.435	.011, .118, .074
RTH Chironomid	RHA	−0.531	.282	.016
	RHA + Hydrology 2	−0.536, 0.279	.359	.013, .169
	RHA + Hydrology 1	−0.471, −0.243	.337	.034, .249
QMH Abundance	RHA + Hydrology 1	−0.426, 0.543	.363	.048, .015
	Hydrology 1	0.439	.192	.053

(Continues)

TABLE 4 (Continued)

Response variable	Best model	Std. coefficient	R ²	p-value
QMH Richness	Geomorphology	−0.598	.357	.005
	HDI + Hydrology 2	−0.582, 0.402	.448	.005, .040
	HDI + Geomorphology	−0.287, −0.432	.412	.226, .075
	Geomorphology + RHA	−0.540, 0.233	.408	.012, .243
QMH Simpsons Diversity	RHA	0.370	.137	.109
	HDI	−0.323	.104	.165
	RHA + Water quality	0.423, 0.303	.226	.068, .180
	HDI + Water quality	−0.386, 0.308	.195	.101, .185
	Water quality	0.228	.052	.333
QMH EPT	RHA	0.440	.291	.014
	RHA + Hydrology 1	0.463, 0.311	.382	.031, .132
	RHA + Hydrology 2	0.545, −0.260	.359	.012, .198
QMH Chironomid	HDI	0.412	.170	.071
	HDI + Geomorphology	0.591, −0.309,	.234	.036, .252

Significant relationships are in bold. HDI, Hydrologic Disturbance Index; RHA, Rapid Habitat Assessment score.

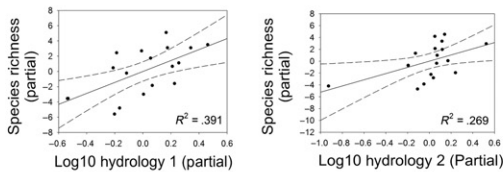
**FIGURE 3** Significant fish, crayfish and macroinvertebrate–environment relationships in 2012. Dashed lines represent 95% confidence intervals

diversity are positively associated with increasing habitat complexity, whereas other studies (Closs & Lake, 1996; Herbert & Gelwick, 2003) indicate that local hydrologic disturbances, such as those reflected at sites with lower RHA scores, can strongly impact fish community dynamics. In 2013 (flood year), hydrology was more strongly related to fish species richness and accounted for a significant portion of the variability in many of the best multiple regression models. Additionally, variables relating to flood frequency and flow variability were particularly important with respect to the Hydrology

1 and Hydrology 2 axes in 2012 (Figure 5), supporting the findings of other investigators who showed that flood frequency has an important influence on aquatic communities in the Ozark Highlands (Dodds, Gido, Whiles, Fritz, & Matthews, 2004; Matthews, Marsh-Matthews, Cashner, & Gelwick, 2013; Matthews et al., 2014).

The negative relationship between habitat disturbance and fish diversity during the drought year but not during the flood year may indicate that droughts exacerbate the effects of anthropogenic disturbances on fish assemblages. In a review of the effects of altered

Fish



Macroinvertebrates

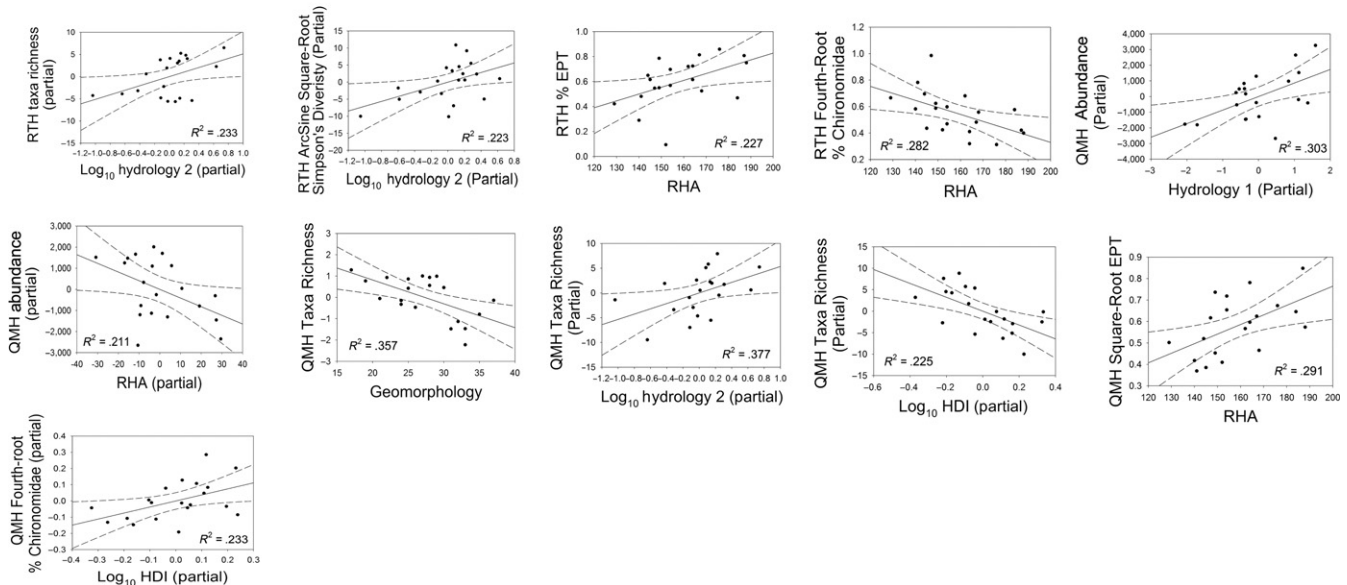


FIGURE 4 Significant fish and macroinvertebrate–environment relationships in 2013. Dashed lines represent 95% confidence intervals

flow regimes on aquatic biodiversity, Bunn and Arthington (2002) showed that streams with altered hydrologic regimes are often associated with reduced diversity of aquatic organisms, as demonstrated for fish (Copp, 1990; Stanford & Ward, 1986) as well as macroinvertebrates (Munn & Brusven, 2004). Falcone et al. (2010) HDI incorporates land-use metrics in addition to direct hydrologic impacts at the catchment scale; the relationship between aquatic biodiversity and land use has also been demonstrated in a number of studies for differing taxonomic groups (Allan, 2004; Walser & Bart, 1999; Wang, Lyons, & Kanehl, 2001).

An important caveat to conclusions drawn about relationships involving fish species richness or diversity in this study is that sites spanned three different river basins, resulting in a somewhat different pool of potential species. While some of the differences in species composition are ecologically trivial, consisting of pairs of closely related allopatric species, others may be more significant. A number of fishes found in the White River basin are not found at Arkansas River basin sites (Robison & Buchanan, 1988). While less than ideal, selecting sites within several major basins present in our study area allowed us to sample at a considerably larger number of sites than we could have had we restricted efforts to a single basin.

One surprising finding was the positive association identified between Water Quality PCA Axis 1 (which loaded positively on Total N, Total P, and conductivity) and crayfish diversity in 2012. This appears to be contrary to what would be expected based on a study

of other taxonomic groups in similar streams (Evans-White, Dodds, Huggins, & Baker, 2009). While this relationship was unexpected, a possible explanation for why water quality in general was more important in 2012 than 2013 could be the general understanding that biota tend to be concentrated at higher densities in smaller suitable habitats during drought periods. Under these conditions, they may experience harsher biotic and abiotic stressors than those to which they are typically exposed (Magoulick & Kobza, 2003). Under such conditions, water quality could potentially become more important than hydrology or catchment-scale variables.

A significant positive relationship was found between geomorphology and per cent generalist crayfish in 2012. Geomorphology PCA Axis 1 loaded positively on erosion and negatively on various measures of substrate size in 2012. Erosion is frequently associated with more disturbed habitats and changes in substrate composition could be related to increased siltation also associated with disturbance (Ward, Tockner, Arscott, & Claret, 2002), which may indicate that disturbed habitats could facilitate the proliferation of these species. In addition to being habitat generalists that are likely more tolerant of disturbance (Larson et al., 2009), both of the species that met this designation are highly fecund, which can offset effects of more disturbed ecological conditions (Carlisle et al., 2010).

Qualitative multihabitat (QMH) macroinvertebrate richness and QMH EPT was, on average, higher in 2013 (flood year) than 2012 (drought year), which supports the findings of previous studies (e.g.

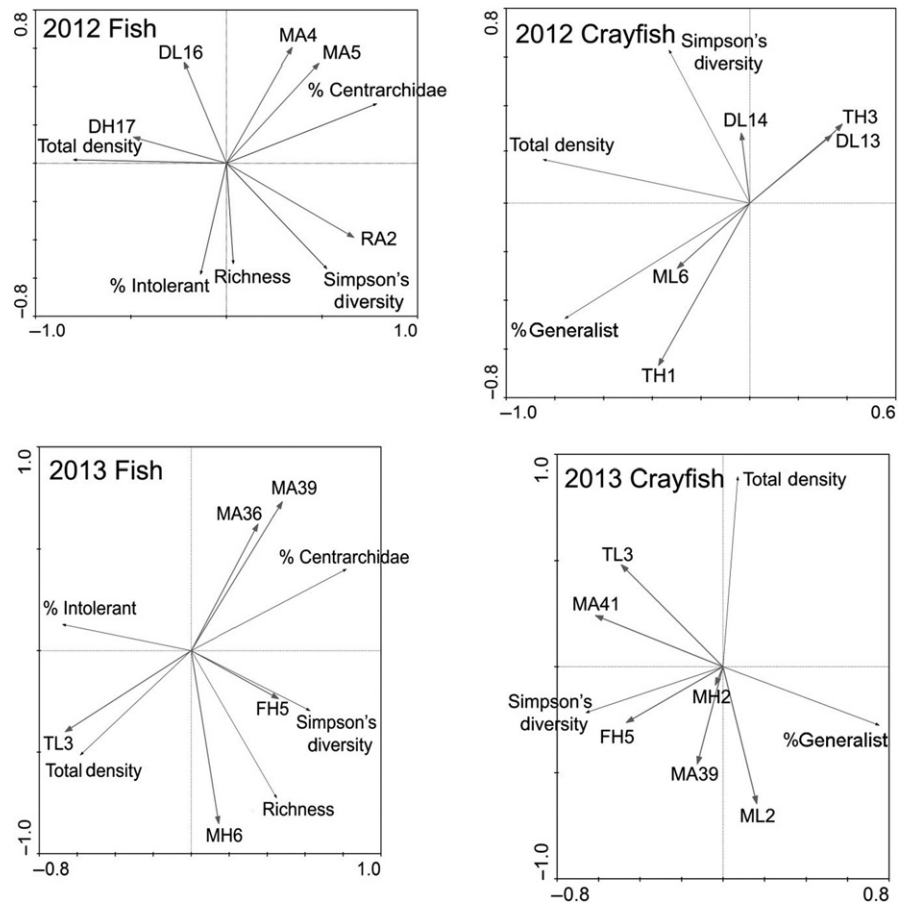


FIGURE 5 Redundancy analysis ordination plots relating fish and crayfish assemblages to selected hydrologic metrics in 2012 and 2013. Angles of arrows indicate associations, and length of arrows indicates strength of the relationship

Lake, 2000, 2003; Suren & Jowett, 2006) which emphasised the effects of drought versus high flows on freshwater communities. RHA was identified as a significant indicator in QMH EPT taxa in both 2012 and 2013. This may indicate that reach-scale habitat structure and quality plays a consistently important role in supporting taxa condition, at least qualitatively, during high and low-flow years. We saw a similar pattern for RTH taxa richness and fish species richness between years; however, geomorphology tended to be most important during the drought year whereas hydrology was most important during the flood year. The positive relationship identified between Hydrology 2 and RTH macroinvertebrate diversity and richness in 2013 emphasises the importance of flow variability and flood frequency for supporting the long-term integrity of aquatic communities.

Our results are consistent with findings of other investigators who showed that chironomid richness increases and EPT taxa richness decreases with increased perturbation (Barbour et al., 1999; Karr, 1991). Both HDI and RHA account for numerous characteristics of ecological disturbance on two different spatial scales (catchment and reach scale, respectively); our results demonstrate the usefulness of macroinvertebrate community response variables as indicators of ecological condition at differing spatial scales within these systems.

Overall, there were many more significant relationships found for the macroinvertebrate assemblage in 2013 than in 2012.

Stubbington, Boulton, Little, and Wood (2015) found that diversity of benthic macroinvertebrates decreased during a major drought and that assemblages at individual streams became more similar, reflecting lower diversity and abundance overall. This increased homogeneity across sites may make relationships between the macroinvertebrate community and drought effects more difficult to detect.

4.1 | Redundancy analysis

The relative importance of flow components was fairly consistent between fish and crayfish. Many magnitude, frequency and duration metrics were directly and inversely related to the fish and crayfish assemblages (Figure 5). While few studies have directly compared the relative importance of flow metric categories, regional environmental flow studies in recent years have indicated that changes in the magnitude of flow may be one of the most important influences on aquatic communities (Armstrong, Richards, & Levin, 2011; Kendy et al., 2012; Monk et al., 2006).

In 2012 (drought year), fish species richness and per cent intolerant species were negatively related to measures of flow variability and duration of low flows. Poff and Allan (1995) showed that stable streams are characterised by the presence of intolerant and specialised species that are often not found in streams with more hydrologic variability. In contrast, per cent Centrarchidae was found

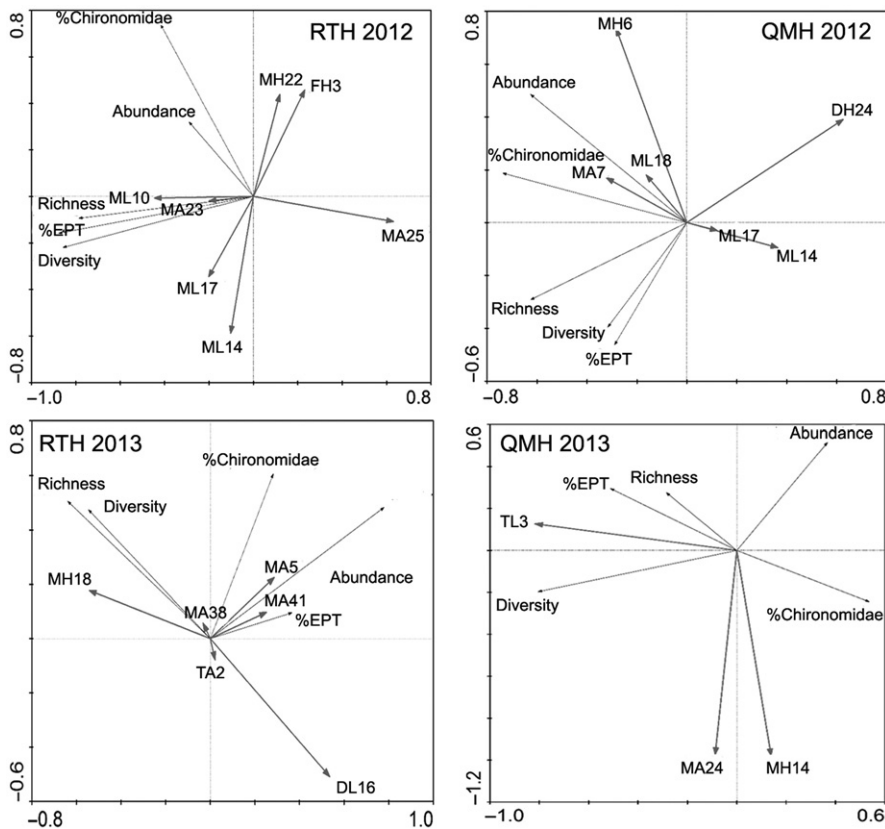


FIGURE 6 Redundancy analysis ordination plots relating RTH and QMH macroinvertebrate assemblages to selected hydrologic metrics in 2012 and 2013. Angles of arrows indicate associations, and length of arrows indicates strength of the relationship

to be positively related to flow variability. A possible explanation may be because centrarchid species tend to prefer pools, more stable habitats during both low and high flow events compared to riffles and runs (Magoulick & Kobza, 2003); that this could somewhat ameliorate the effects of increased flow variability on these species as compared to other groups or guilds (Poff et al., 2010). Based on the ordination plots for 2012, the timing of flows (e.g. TH1 and TH3) appeared to be more important for crayfish than for the fish assemblages. This is most apparent for % generalist crayfish; less predictable timing of floods and floods later in the year seem to favour these species, perhaps because of life history traits (Carlisle et al., 2010; Larson & Olden, 2010; Pflieger, 1996). No individual flow variable in 2012 was important to both fish and crayfish communities, which illustrates why flow can differentially affect taxonomic assemblages in the same region.

In 2013, there were more flow variables in common between fish and crayfish assemblages than there were in 2012. Three metrics, MA39 (variation in monthly flows), FH5 (flood frequency) and TL3 (predictability of low flows) were important to both groups and showed similar relationships. In both groups, monthly flow variation was strongly negatively related to total density, predictability of low flows was positively related to total density, and flood frequency was closely positively related to diversity. Fishes and other aquatic fauna in Ozark streams are typically well adapted to cope with floods, even those considered catastrophic (Dodds et al., 2004; Matthews et al., 2014), which is why the strong positive relationship found between FH5 and fish and crayfish diversity is not surprising.

Flooding may generally have less of a detrimental effect than drought in riverine ecosystems (Lake, 2000; Matthews et al., 2013). In long-term studies, Matthews et al. (2013) found that frequency and magnitude of flooding events are important in supporting the structure and function of stream fish communities. It is possible that frequent flooding may help facilitate greater diversity in Ozark Highland streams.

Magnitude was the most influential flow component for macroinvertebrate communities in Groundwater Flashy streams in the Ozark Highlands (Figure 6). Other flow components (e.g. timing, frequency and duration) did exert some influence, but to a much lesser extent. This supports what others have found with respect to flow magnitude being a particularly influential component on the biological integrity of aquatic communities compared to other chemical, physical and hydrologic covariates (Armstrong et al., 2011; Kendy et al., 2012).

In 2012, RTH macroinvertebrate assemblage richness, diversity, and % EPT were strongly and positively related to average and low flow magnitude metrics (MA23 and ML10, respectively) and negatively related to the coefficient in variation of February flows (MA25). The close association between flow variables and RTH macroinvertebrate metrics compared to the QMH is likely related to the sampling method, as EPT taxa are typically associated with riffles and could account for the higher EPT taxa richness and diversity found in that habitat (Gregory, 2007).

While we found some consistent relationships among the RTH and QMH analyses (e.g. the overall importance of magnitude

metrics), the important flow variables differed markedly between sampling types and years. Interestingly, hydrologic metrics related to low flow were more important during the drought year (2012), whereas average flow metrics tended to be more important in the flood year (2013) and high flow metrics were important in both years. Because these are flashy streams, periodic flooding and high flow events may be important to supporting biotic integrity in these systems. Further research will be necessary to better understand the mechanistic complexities embedded in these flow-ecology relationships.

It is important to note that these predicted natural flow metrics are based on a consistent period of record and do not actually change between the two years; it is the invertebrate communities themselves that strongly differ between one year and the next and between one sampling method and the other. These shifts, however, are enough to change the relationships between response variable and flow metrics. While it has been a traditional tenet of long-term biomonitoring programmes that reference-quality streams show relatively little interannual variation in macroinvertebrate communities (Robinson, Minshall, & Royer, 2000), recent studies have shown that this is not always the case. MacDonald and Cote (2014) compared year-to-year variation in benthic macroinvertebrate stream communities at reference and urbanised sites over a six year period and found significant temporal change in communities in both stream types, although there was a much a greater degree of variation at urbanised sites. Our study sites encompassed a gradient of sites ranging from highly disturbed to reference condition. Carter and Fend (2001) found that strong interannual differences in discharge regimes, similar to those seen in our study, resulted in strong interannual variation in percentages of both taxonomic groups and functional feeding groups among macroinvertebrate communities.

The notable differences in ecological response variables associated with fish, crayfish and macroinvertebrate communities in this study between two years under different environmental conditions (drought versus flood) may indicate that in order to form a strong picture of relationships between biota and hydrologic variables, long-term monitoring over multiple years encompassing a variety of environmental extremes may be crucial. Had our study taken place during two consecutive drought summers, for example, we likely would have found more consistent results between the two years. However, flow-ecology relationships underpinned by such a dataset would be missing the critical interyear variation in aquatic assemblages evaluated as part of this study. ELOHA concepts are increasingly being applied to the study and management of freshwater streams in regions across the world: in North America (Buchanan, Moltz, Haywood, Planner, & Griggs, 2013; McManamay, Orth, Doll-off, & Matthews, 2013; Novak et al., 2016; Reidy Liermann et al., 2012; Sanderson et al., 2012), Europe (Belmar, Velasco, & Martinez-Capel, 2011), Asia (Zhang et al., 2012), Australia (Rolls & Arthington, 2014) and Africa (O'Brien et al., 2017). While the implementation of these principals is leading to a paradigm shift in our understanding of freshwater ecology, it could be improved by the incorporation of a better understanding of spatial and temporal variation, including

more infrequent and extreme disturbances (Sousa, 1984; Stoddard et al., 2006), as suggested by this study.

5 | CONCLUSIONS

Environment-ecology relationships in this study showed substantial differences between years (drought versus flood) and across taxonomic groups. Using biological data to inform policies and models can improve the likelihood of achieving desired results (King et al., 2015; Poff et al., 2010), yet there is a great deal of variation inherent in the collection of such data (Hurlbert, 1984; Ruiz-Gutierrez, Hooten, & Grant, 2016). While long-term monitoring studies designed to assess year-to-year temporal variation at the same site would be the ideal way to better understand environmental flow relationships (MacDonald & Cote, 2014), limitations of time and funding make this an impractical option. Most studies examining flow-ecology relationships are carried out over a shorter time frame, and some studies are based on a single sample from a given time period to categorise a site (Olden et al., 2014). Understanding temporal variation in flow-ecology relationships could be especially important for resource conservation and management. Not all relationships observed in this study showed interyear variation. Some relationships, for example, the positive relationship between RHA and EPT taxa, were temporally stable. The ability to identify and focus on temporally stable relationships, or to better account for the variability in those that do show strong year-to-year variation, could be of great value in the development of an improved framework for incorporating environmental flows into the management arena. We also saw differences in which metrics influenced different taxonomic assemblages and the ways in which they were affected. While this kind of complexity poses a substantial challenge in terms of conservation, understanding which variables are consistently important across assemblages, such as flood frequency or flow variability, can provide a focus for management efforts.

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